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MILITARY SPECIFICATION

VALVE, FLUID INJECTION, SOLENOID OPERATED

This specification is approved for use by the U.S. Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification defines the requirements for one type of solenoid operated fluid injection valve.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standard, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Missile Command, ATTN: AMSMI-RD-SE-TD-ST, Redstone Arsenal, AL 35898-5270 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

L-P-378	-	Plastic, Sheet and Strip, Thin Gauge, Polyolefin
QQ-P-35	-	Passivation Treatments for Corrosion-Resistant Steel
QQ-S-763	-	Steel Bars, Wire, Shapes, and Forgings, Corrosion Resisting
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-C-1797	-	Cushioning Material, Resilient, Low Density, Unicellular, Polypropylene Foam
PPP-T-76	-	Tape, Packaging, Paper (for Carton Sealing).

MILITARY

MIL-P-116	-	Preservation, Methods of
MIL-B-117	-	Bags, Sleeves and Tubing
MIL-T-152	-	Treatment, Moisture- and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment
MIL-S-7742	-	Screw Threads, Standard, Optimum Selected Series: General Specification for
MIL-I-23053/5	-	Insulation Sleeving, Electrical, Heat Shrinkable, Polyolefin, Flexible, Crosslinked
MIL-P-25604	-	Propellant, uns-Dimethylhydrazine

STANDARDS

FEDERAL

FED-STD-313	-	Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities

MILITARY

MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-130	-	Identification Marking of U.S. Military Property
MIL-STD-202	-	Test Methods for Electronic and Electrical Component Parts
MIL-STD-461	-	Electromagnetic Emission and Susceptibility Requirements
MIL-STD-462	-	Electromagnetic Interference Characteristics, Measurement of

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MIL-STD-810	-	Environmental Test Methods and Engineering Guidelines
MIL-STD-889	-	Dissimilar Metals
MIL-STD-970	-	Standards and specifications, Order of Preference for the Selection of
MIL-STD-1523	-	Age Controls of Age - Sensitive Elastomeric Material (for Aerospace Applications)
MIL-STD-2000	-	Soldering, Manual Type, High Reliability Electrical and Electronic Equipment
MS3116	-	Connectors, Plug, Electric, Series 1, Solder Type, Straight, Bayonet Coupling
MS20995	-	Wire, Safety or Lock
MS33540	-	Safety Wiring and Cotter Pinning, General Practices for
MS33656	-	Fitting End, Standard Dimensions for Flared Tube Connection and Gasket Seal
MIL-STD-45662	-	Calibration Systems Requirements

(Unless otherwise indicated, copies of the federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Ave., Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.2 and 6.3) in accordance with 4.4.

3.2 Material. Preference shall be given to selection of materials and processes such that a storage life of five years shall be expected. The use of critical or strategic materials shall be kept to a minimum consistent with the design requirements (see 4.6.1). Materials not definitely specified shall be selected in accordance with the order of preference specified in MIL-STD-970.

3.2.1 Dissimilar metals. The use of dissimilar metals that tend to corrode when in contact, shall be avoided. Where the use of such metals cannot be avoided, they shall be insulated from each other as specified in MIL-STD-889.

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3.2.2 Corrosion resistant metals. All metals used shall be corrosion resistant or protected to resist corrosion. Corrosion resistant steels shall be protected in accordance with QQ-P-35.

3.2.3 Use of magnesium. Magnesium shall not be used without prior approval of the contracting activity.

3.2.4 Threaded parts. All threaded parts shall conform to MIL-S-7742 and shall be positively locked or safety wired in the installed position. Safety wire, where used, shall be applied in accordance with MS33540, conforming to MS20995, and shall be covered with insulation sleeving conforming to MIL-I-23053/5, class I.

3.2.5 Soldering. Soldering of electrical connections shall be performed in accordance with MIL-STD-2000, Task F.

3.2.6 Fungus proof materials. Materials which do not support the growth of fungi shall be used. However, if nutrient materials are used, they shall be treated in accordance with MIL-T-152.

3.2.7 Elastomeric parts. When elastomeric parts are used, they shall be compatible with operating fluids specified and all test and manufacturing fluids used. Marking shall be in accordance with MIL-STD-1523. Butyl elastomeric seals are not acceptable.

3.2.8 Lead wires. The coil lead wires shall be size 24 Average Wire Gage (AWG).

3.2.9 High temperature exposure metals. All valve parts which extend past the plane of the mounting surface of the valve and into the injection port of an engine shall be of corrosion resistant steel conforming to QQ-S-763, type 347, condition A, or a steel with equivalent high temperature properties: melting point, specific heat, thermal conductivity, and density.

3.3 Physical properties.

3.3.1 Design and construction. The design and construction of the valve shall be in accordance with figure 1 (see 4.6.1).

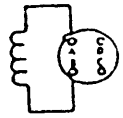
3.3.2 Weight. The dry weight of the valve shall be not greater than 1.1 pounds.

3.3.3 Alignment. When seated, all plungers, poppets, balls and pistons shall not be misaligned and shall not chatter.

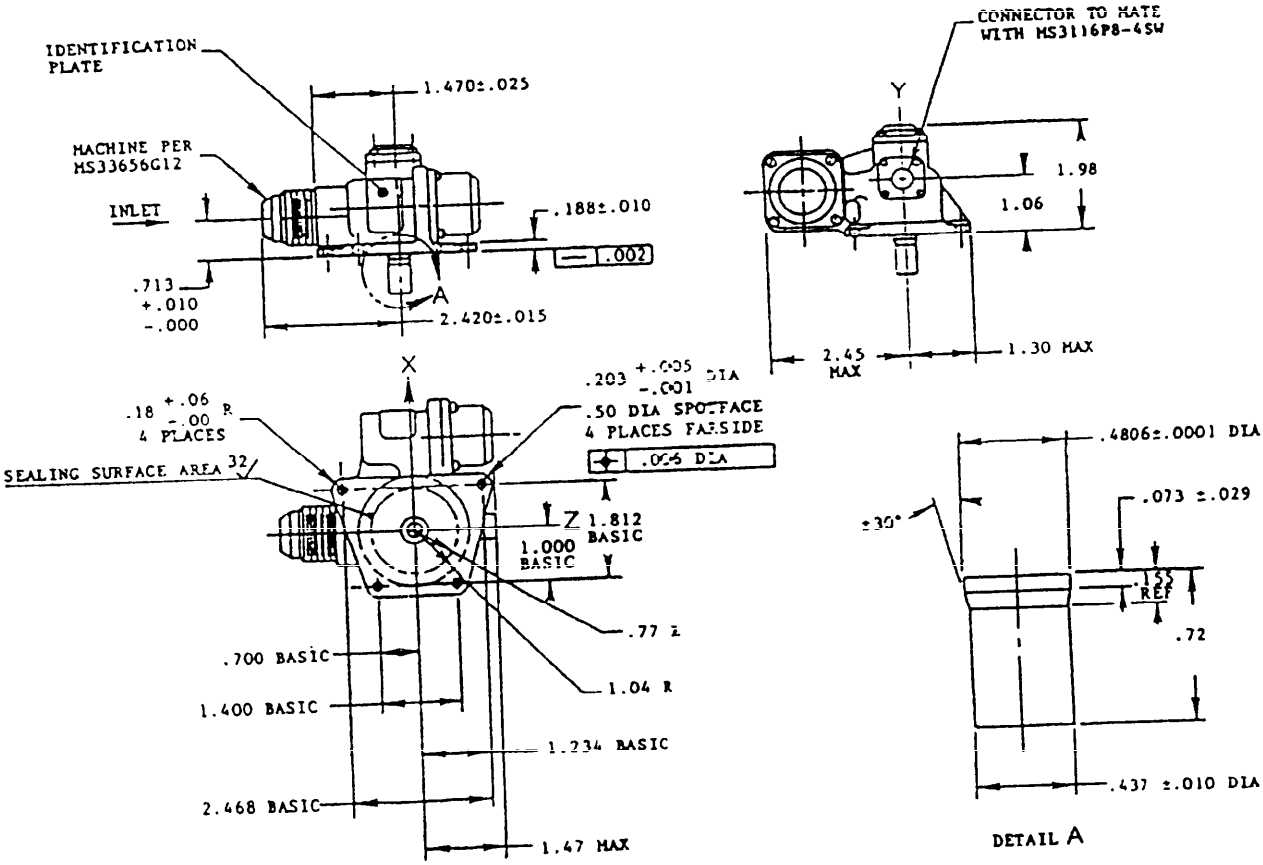
3.3.4 Valve operation. Valve on-off operation shall be accomplished by energizing a solenoid. The initial position of the solenoid shall be off.

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VALVE AXIS	MISSILE AXIS
X	LONGITUDINAL
Y	RADIAL
Z	TRANSVERSE



WIRING DIAGRAM
(NORMALLY CLOSED WHEN DE-ENERGIZED)



NOTES:

1. All dimensions are in inches.
2. Metric equivalents shall be acceptable.

FIGURE 1. Valve, fluid injection, solenoid operated.

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3.3.5 Operating fluid. The valve operating fluid shall be unsynmetrical dimethylhydrazine (UDMH) conforming to MIL-P-25604. All volumetric flow rates are based on MIL-P-25604 fuel usage. An approved test fluid may be used for performance and evaluation unless otherwise specified.

WARNING

UDMH is a hazardous material and should be handled in accordance with the manufacturer's safety data sheet.

3.3.5.1 Hazardous material. UDMH has been determined to be corrosive, toxic, and hazardous. Markings shall be as specified (see 5.3 and 6.2). Contractors shall submit Material Safety Data Sheets in accordance with FED-STD-313 as specified in the contract or order (see 6.2). The following statement shall be included:

"The material shall have no adverse effect on the health of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred by the contracting activity to the appropriate departmental medical service who will act as an advisor to the contracting agency."

3.3.5.2 Fluid contamination. The valve shall perform as specified when the operating fluid is contaminated to the levels shown in table I.

TABLE I. Level of contaminants.

Particle size, microns	Particles per 100 milliliters 1/
<10	Unlimited
10-25	6000
26-50	2000
51-100	800
101-300	50
301-500	5
>500	0

1/ Maximum contaminants = 15 milligrams per liter.

3.3.5.3 Flow rate. The valve shall flow 5.5 pounds per second (lb/sec) minimum and 5.7 lb/sec maximum of UDMH with a 900 ± 10 pound per square inch (psi) pressure drop across the valve (see 4.6.2).

3.3.5.4 Flow stability. The flow stability of the valve shall be such that after 1000 cycles of operation, the injectant flow rate with 900 ± 10 psi across the valve and a regulated four hertz (Hz) square wave applied for 230 seconds, shall be within three percent of the injectant flow rate measured after 100 cycles of operation (see 4.6.11).

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3.3.6 Normal operating pressure. The valve shall operate as specified at 900 to 1100 pounds per square inch gage (psig) (see 4.6.3.1).

3.3.6.1 Surge pressure. The valve shall withstand pressure surges from 0 to 2300 psig peak for a period which shall be not less than 10 milliseconds and not greater than 100 milliseconds with no degradation of performance (see 4.6.3.2).

3.3.6.2 Proof pressure. The valve in the de-energized condition shall withstand a proof pressure of 1650 psig for 3 minutes. There shall be no external leakage, rupture, permanent deformation or degradation of performance (see 4.6.3.3).

3.3.6.3 Burst pressure. The valve in the de-energized condition shall withstand a burst pressure of 4500 psig for 3 minutes. There shall be no external leakage (see 4.6.3.4).

3.3.6.4 Initial pressurization operating characteristics. The valve full opening and full closing times at specified pressure levels shall be as shown in table II (see 4.6.4).

TABLE II. Pressure vs. full opening and closing times.

Pressure across valve (psi)	Opening and closing times (milliseconds)
400	15 ± 6
800	13 ± 4
900	See table III

3.3.6.5 Operating temperature. The valve shall perform as specified when installed on a small liquid propellant rocket engine which is ignited and fired for a maximum duration of 6.5 seconds. The heat added to the valve shall be in addition to the initial temperature conditions of the valve and UDMH of -40 to 72° Celsius (C) (see 4.6.5).

3.3.6.6 Internal leakage. The internal leakage shall be not greater than 1.0 gallons per minute (gpm) of UDMH with the valve de-energized and with 1000 psig applied to the inlet (see 4.6.6).

3.3.6.7 External leakage. The external leakage shall be zero at any applied pressure not greater than 4500 psig (see 4.6.7).

3.3.6.8 Injection stream shape. The injection stream shape, injected into the atmosphere with 900 ± 10 psi across the valve, shall have a total included angle not greater than 13° at a location 4 ± 0.004 inches downstream from the surface of the valve mounting flange (see 4.6.8).

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TABLE III. Valve operating time (milliseconds).

Temperature	Full on			Full off		
	Electrical delay 1/	Output flow rise time 1/	Total	Electrical delay 1/	Output flow decay time 1/	Total
16 ± 33°C	7 ± 2.5	4 ± 2	12 ± 3	7.5 ± 25	4 ± 2	11 ± 3
At -40°C and 72°C	$7^{+3.5}_{-2.5}$	4 ± 2	12^{+4}_{-3}	7.5^{+3}_{-2}	4 ± 2	11^{+4}_{-3}

1/ Electrical delay, flow rise, and decay times apply only to UDMH usage at the system test level. The total ON and OFF times apply to both UDMH system level usage and UDMH or UDMH simulation fluid usage at the component level.

3.3.6.9 Injection stream momentum. The momentum of the injection stream shall be 93 percent of the ideal momentum with a pressure drop of 900 ± 10 psi across the valve (see 4.6.9).

3.3.6.10 Injection stream perpendicularity. The valve shall be tested as specified in 4.6.10.

3.3.6.11 Operating time and operating life. Valve operating time across the valve shall be 900 ± 10 psi. The valve operating life shall be not less than 10,000 cycles without malfunction (see 4.6.12 and table III).

3.4 Electrical characteristics.

3.4.1 Input voltage and current. Input voltage to the solenoid shall be 25.0 to 33.0 Volts direct current (Vdc) (see 4.6.13.1). The valve shall operate when energized with input currents from 0.551 to 1.27 amperes (A) (see 4.6.13.2).

3.4.2 Coil resistance. The solenoid shall have the temperature versus resistance characteristics shown in table IV (see 4.6.13.3).

3.4.3 Inductance. The inductance of the solenoid shall be 0.85 to 0.88 Henries (H) over the temperature range of -40° to 72°C at frequencies of 0 to 22 Hz (see 4.6.13.4).

3.4.4 Insulation resistance. The insulation resistance between electrical circuits and the case shall be not less than 50 megohms at 500 Vdc for 60 seconds over all operational temperatures (see 4.6.13.5).

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TABLE IV. Temperature vs. resistance characteristics.

Temperature	DC resistance ohm
20°C	36 ± 1.8
72°C	45.4 maximum
-40°C	26.0 minimum

3.5 Environmental. The valve shall be subjected to non-operating and operating environmental testing as specified in 4.6.14.

3.5.1 Non-operating environmental testing.

3.5.1.1 Vibration. The valve, empty of fluid, shall withstand sinusoidal vibration from 10 to 28 Hz at 0.25 inch double amplitude (in da) and 28 to 500 Hz at 10 gravity units (g) peak for six hours in each of the three mutually perpendicular axes as specified in figure 1 (see 4.6.14.1.1).

3.5.1.2 Shock. The valve shall withstand three half-sine shock pulses of 450g applied for three milliseconds in each of three mutually perpendicular axes as specified in figure 1 (see 4.6.14.1.2).

3.5.1.3 Altitude. The valve shall withstand exposure to any altitude from sea level to 50,000 feet for 12 hours (see 4.6.14.1.3).

3.5.1.4 Humidity. The valve shall withstand a stabilized relative humidity of 30 percent at a temperature of 21.2° ± 1.4°C, (see 4.6.14.1.4).

3.5.1.5 Fungus. The valve shall be nonnutritive to fungus growth such as encountered in tropical climates (see 4.6.14.1.5).

3.5.1.6 Storage temperature. The valve shall withstand storage temperatures from -54° to 72°C (see 4.6.14.1.6).

3.5.2 Operating environmental testing.

3.5.2.1 Vibration. The valve shall withstand random vibration of 6.7 g^2/Hz from 50 to 1500 Hz attenuated at 24 decibels per octave on both high and low frequency ends within the overall bandwith of 20 to 2000 Hz with a minimum root mean square (rms) acceleration for this spectrum of 100g for 6.5 seconds along each of the three mutually perpendicular axes as specified in figure 1 (see 4.6.14.2.1).

3.5.2.2 Shock. The valve shall withstand half-sine shock pulses of 600g peak for 1.0 to 1.5 milliseconds in both plus and minus directions of each of the three mutually perpendicular axes as specified in figure 1 (see 4.6.14.2.2).

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3.5.2.3 Altitude. The valve shall withstand exposure to any altitude from sea level to 50,000 feet (see 4.6.14.2.3).

3.5.2.4 Explosive atmosphere. The valve shall not cause ignition when operated in an ambient atmosphere of an explosive gaseous mixture (see 4.6.14.2.4).

3.5.2.5 Boost acceleration. The valve shall withstand acceleration levels applied for 6.5 seconds in accordance with the following (see 4.6.14.2.5):

a. Forward at 40.0g
b. Aft at 3.0g
c. Lateral and up at $\pm$ 2.0g
d. Angular about spin axes at 13.5 radians/sec^2
e. Angular about transverse axes at $\pm$ 2.1 radians/sec^2 at the transverse moment arm of 107 inches.

3.5.2.6 Spin. The valve shall withstand any spin from 0 to 5.5 revolutions per second (rps) at a moment arm of 6.5 inches (see 4.6.14.2.6).

3.5.2.7 Acoustical noise level. The valve shall withstand sound pressure levels as specified in 4.6.14.2.7 in a reverberant sound field for 6.5 seconds.

3.5.2.8 Temperature. The valve shall withstand environmental and operating fluid temperatures from -40° to 72°C (see 4.6.14.2.8).

3.5.3 Electromagnetic interference. Electrical circuits shall not cause radio interference during operation beyond the limits as specified in MIL-STD-461 and MIL-STD-462 (see 4.6.15).

3.6 Marking. The valve shall be marked in accordance with MIL-STD-130 (see 4.6.1).

3.7 Workmanship. There shall be no cracks, burrs, dents, loose attaching parts or misalignment or any other evidence of poor workmanship that would make the valve unsuitable for the purpose intended (see 4.6.1).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections

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are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all tests shall be conducted under the following conditions:

a. Ambient temperature; 26.7° ± 16.6°C

b. Test fluid temperature; 32.2° ± 11.1°C

c. Relative humidity; 90 percent or less

d. Barometric pressure; 30 ± 2 inches of mercury (in Hg).

4.3.1 Inspection equipment. The quality and performance of the equipment used in conducting the inspections specified shall be compatible with the requirements of inspections to be performed and shall be controlled by a test equipment calibration system in accordance with MIL-STD-45662.

4.4 First article inspection. When specified in the contract or order (see 6.3), a first article sample shall be subjected to first article inspection. Subsequent units shall not be considered for acceptance until Government approval of the first article sample has been obtained.

4.5 Quality conformance inspection. Quality conformance inspection shall be as specified in this paragraph. Production lots shall comply with all of the requirements specified in this paragraph. Failure to successfully complete any of the inspections specified herein constitutes rejection of the lot.

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4.5.1 Inspections. Inspections for acceptance of inspection lots shall be as specified in table V to verify the corresponding requirements of Sections 3 and 5 indicated therein.

TABLE V. Quality conformance inspection.

Characteristics	Requirement paragraph	Method paragraph	Frequency of inspection
Visual examination	3.2.4, 3.2.5, 3.2.8, 3.3, 3.4, 3.5	4.6.1	Table VI
Dimension examination	3.3, 3.3.1	4.6.1	Table VI
Flow rate	3.3.5.3	4.6.2	100 percent
Normal operating pressure	3.3.6	4.6.3.1	100 percent
Surge pressure	3.3.6.1	4.6.3.2	Table VI
Proof pressure	3.3.6.2	4.6.3.3	Table VI
Internal leakage	3.3.6.6	4.6.6	100 percent
External leakage	3.3.6.7	4.6.7	100 percent
Injection stream shape	3.3.6.8	4.6.8	Table VI
Injection stream momentum	3.3.6.9	4.6.9	Table VI
Injection stream perpendicularity	3.3.6.10	4.6.10	Table VI
Operating time and life	3.3.6.11	4.6.12	100 percent
Coil resistance	3.4.2	4.6.13.3	100 percent
Input current	3.4.1	4.6.13.2	100 percent
Inductance	3.4.3	4.6.13.4	Table VI
Insulation resistance	3.4.4	4.6.13.5	Table VI

4.5.2 Lot formation. An inspection lot shall consist of a group like items produced at the same place using the same batches of materials, lots of components, process runs, fabrication techniques, assembly techniques, tools, equipment, and facilities, but shall not exceed one month's production. Upon submittal to the procuring activity for acceptance, each lot shall be accompanied by in-process inspection records and inspection results as required by the procuring activity inspector and this specification.

4.5.3 Sampling. A sample of production items as specified in the contract or order (see 6.2) shall be selected from each lot presented for Government acceptance in accordance with table VI. The sample shall be inspected in accordance with table V to verify the corresponding requirements of Sections 3 and 5. All characteristics are designated Major and any discovered defect shall constitute rejection of the represented lot.

4.5.4 Inspection of packaging. Preservation, packing, and marking shall be inspected for conformance to section 5.

4.6 Methods of inspection.

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TABLE VI. Sampling plan.

Lot Size	Sample Size
2 to 13	100%
14 to 150	13
151 to 280	20
281 to 500	29
501 to 1200	34
1201 to 3200	42
3201 to 10,000	50
10,001 to 35,000	60
35,001 to 150,000	74
150,001 to 500,000	90
500,001 and over	102
In all cases:	Acceptance number is ZERO Rejection number is ONE

4.6.1 Examination of product. Examination of materials and processes used in the manufacture of the valve shall be made to determine conformance to section 3 (see 3.6 and 3.7) requirements. The dimension and tolerance requirements of this specification shall be verified to determine conformance with figure I and 3.3.1.

4.6.2 Flow rate. The valve shall be cycled four times from zero to full current at 0.25 Hz and the flow rate measured for conformance to 3.3.5.3.

4.6.3 Pressure ratings.

4.6.3.1 Normal operating pressure. The valve shall be pressurized and tested for conformance to 3.3.6.

4.6.3.2 Surge pressure. The valve shall be pressurized and tested for conformance to 3.3.6.1.

4.6.3.3 Proof pressure. The valve shall be pressurized to proof pressure and inspected for conformance to 3.3.6.2.

4.6.3.4 Burst pressure. The valve shall be pressurized to burst pressure and inspected for conformance to 3.3.6.3.

4.6.4 Initial pressurization operating characteristics. The valve shall be pressurized and the full opening and full closing times measured for conformance to 3.3.6.4.

4.6.5 Operating temperature. The valve shall be installed on a small liquid propellant rocket engine and tested for conformance to 3.3.6.5. The

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quantity of valves to be tested and the quantity of engines required shall be as specified in the contract or order (see 6.2).

4.6.6 Internal leakage. The valve shall be pressurized and tested for conformance to 3.3.6.6.

4.6.7 External leakage. The valve shall be monitored for external leakage during all functional testing (see 3.3.6.7).

4.6.8 Injection stream shape. The valve shall be tested for conformance to 3.3.6.8.

4.6.9 Injection stream momentum. The valve shall be tested for conformance to 3.3.6.9. The momentum of the injection stream shall be measured on a circular force plate 1.2715 ± 0.0005 inches in diameter located 4.000 ± 0.004 inches from and perpendicular to the surface of the valve mounting flange directly in line with the centerline of the flow metering pin.

4.6.10 Injection stream perpendicularity. The centerline of the injection stream deviation shall be not greater than 0.5° from a line drawn perpendicular to the surface of the valve mounting flange and passing through the centerline of the flow metering pin, when the center of the load cell force plate is not greater than 0.035 inch from the true center of the flow metering pin (see 3.3.6.10).

4.6.11 Flow stability. The valve flow stability shall be tested for conformance to 3.3.5.4.

4.6.12 Operating life. The operating life test shall be a continuation of the flow stability test. The input signal shall be regulated to provide a 33 ± 2 Hz square wave signal to cycle the valve for conformance to 3.3.6.11.

4.6.13 Electrical characteristics.

4.6.13.1 Input operational voltage. The valve shall be tested for conformance to 3.4.1.

4.6.13.2 Input current. The valve shall be energized and tested for conformance to 3.4.1.

4.6.13.3 Coil resistance. The coil resistance across pins A and B of the electrical connector shall be measured in accordance with MIL-STD-202, method 303 for conformance to 3.4.2.

4.6.13.4 Inductance. The inductance of the valve coil shall be measured for conformance to 3.4.3.

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4.6.13.5 Insulation resistance. The insulation resistance of the valve coil shall be measured in accordance with MIL-STD-202, method 302, test condition B for conformance to 3.4.4.

4.6.14 Environmental. The valve shall meet the requirements of this specification after being subjected to the non-operating environmental conditions and while being subjected to the operating environmental conditions or any natural combination thereof (see 3.5).

4.6.14.1 Non-operating environmental testing.

4.6.14.1.1 Vibration. The valve shall be tested in accordance with MIL-STD-810, method 514.4 for conformance to 3.5.1.1.

4.6.14.1.2 Shock. The valve shall be tested in accordance with MIL-STD-810, method 516.4 for conformance to 3.5.1.2.

4.6.14.1.3 Altitude. The valve shall be tested in accordance with MIL-STD-810, method 500.3 procedure I for conformance to 3.5.1.3.

4.6.14.1.4 Humidity. The valve shall be tested in accordance with MIL-STD-810, method 507.3, procedure II for conformance to 3.5.1.4.

4.6.14.1.5 Fungus. The valve shall be tested in accordance with MIL-STD-810, method 508.4 for conformance to 3.5.1.5.

4.6.14.1.6 Storage temperature.

a. High temperature. The valve shall be tested in accordance with MIL-STD-810, method 501.3, procedure I for conformance to 3.5.1.6.

b. Low temperature. The valve shall be tested in accordance with MIL-STD-810, method 502.3 procedure I for conformance to 3.5.1.6.

4.6.14.2 Operating environmental testing.

4.6.14.2.1 Vibration. The valve shall be tested in accordance with MIL-STD-810, Method 514.4 for conformance to 3.5.2.1.

4.6.14.2.2 Shock. The valve shall be tested in accordance with MIL-STD-810, method 516.4 procedure IV for conformance to 3.5.2.2.

4.6.14.2.3 Altitude. The valve shall be tested in accordance with MIL-STD-810, method 500.3, procedure II for conformance to 3.5.2.3.

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4.6.14.2.4 Explosive atmosphere. The valve shall be tested in accordance with MIL-STD-810, method 511.3, procedure I for conformance to 3.5.2.4.

4.6.14.2.5 Boost acceleration. The valve shall be tested in accordance with MIL-STD-810, method 513.4 procedure II for conformance to 3.5.2.5.

4.6.14.2.6 Spin. The valve shall be tested in accordance with instructions for spin testing provided by the contracting officer (see 6.2).

4.6.14.2.7 Acoustical noise level. The valve shall be tested in accordance with MIL-STD-810, method 515.4 for conformance to 3.5.2.7.

4.6.14.2.8 Temperature.

a. High temperature. The valve shall be tested in accordance with MIL-STD-810, method 501.3, procedure II for conformance to 3.5.2.8.

b. Low temperature. The valve shall be tested in accordance with MIL-STD-810, method 502.3, procedure II for conformance to 3.5.2.8.

4.6.15 Electromagnetic interference. The valve shall be tested in accordance with MIL-STD-461 and MIL-STD-462, for conformance to 3.5.3.

5. PACKAGING

5.1 Preservation. Preservation shall be level A.

5.1.1 Cleaning. Valves shall be cleaned so that the contamination added by the valve to the purge fluid, when 100 milliliters of purge fluid is passed through the unit under pressure, shall be not greater than the limits of table I. The purge fluid shall be compatible with all materials used in the construction of the unit and the operating fluid.

5.1.2 Drying. Drying shall be in accordance with MIL-P-116.

5.1.3 Preservative application. Preservatives shall not be used.

5.1.4 Unit packs. Each valve shall be individually unit packed in contamination barrier material of polyethylene plastic film conforming to L-P-378 or polyethylene plastic bags conforming to MIL-B-117. The film or bags shall be closed by either heat sealing or with tape.

5.2 Packing. Packing shall be level A.

5.2.1 Level A. Valves preserved as specified in 5.1 shall be packed in containers conforming to PPP-B-636, style RSC, type CF, grade 175, class DOM

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and cushioned with polypropylene foam material conforming to PPP-C-1797. Container closure shall be in accordance with PPP-T-76 with barrier in accordance with MIL-B-117.

5.3 Marking.

5.3.1 Unit packs. Valves which have been preserved as specified in 5.1 shall have a label attached which contains, as a minimum, the following:

CAUTION. This item has been specially cleaned and sealed in a contamination barrier. Do not remove contamination barrier until ready for installation.

5.3.2 Exterior containers. In addition to any special or other identification marking required by the procuring activity (see 6.2), each container shall be marked in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The valve is intended for use in a small liquid propellant rocket engine to regulate the UDMH fuel injection in the secondary thrust vector control. The valve shall provide rapid on-off control of liquid propellant under pressure that is injected into an engine thrust chamber.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).

c. When first article is required (see 3.1)

d. Sampling plan to be used for quality conformance (see 4.5.2).

e. Quantity of valves and engines required (see 4.6.5).

f. Applicable level of preservation, packing and marking if other than specified (see section 5).

g. Requirements for valve spin testing (see 4.6.15.2.6).

h. Material Safety Data Sheets (see 3.3.5).

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6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerers whether the item(s) should be a first article sample, a first production item, or a number of items to be tested as specified in 4.3. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Metrication. Metric equivalents in accordance with FED-STD-376 are acceptable for use in this specification.

6.5 Subject term (keyword) listing.

Carrier, liquid rocket fuel
Mechanical plumbing
Pump
Shutoff

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - MI

Review activities:
DLA - CS

Preparing activity:
Army - MI

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-V-47320A(MI)	14 December 1990

3. DOCUMENT TITLE
VALVE, FLUID INJECTION, SOLENOID OPERATED

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER		
a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
COMMANDER U.S. ARMY MISSILE COMMAND	(1) Commercial (205) 876-6980	(2) AUTOVON 746-6980
c. ADDRESS (Include Zip Code) ATTN: AMSMI-RD-SE-TD-ST REDSTONE ARSENAL, AL 35898-5270	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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